

1. Binary Search Easy Topics premium lock icon Companies Given an array of integers nums which is sorted in ascending order, and an integer target, write a function to search target in nums. If target exists, then return its index. Otherwise, return -1.

You must write an algorithm with  $O(\log n)$  runtime complexity.

Example 1:

Input: nums = [-1,0,3,5,9,12], target = 9 Output: 4 Explanation: 9 exists in nums and its index is 4

Example 2:

Input: nums = [-1,0,3,5,9,12], target = 2 Output: -1 Explanation: 2 does not exist in nums so return -1

Constraints:

$1 \leq \text{nums.length} \leq 10^4$

$-10^4 < \text{nums}[i], \text{target} < 10^4$

All the integers in nums are unique. nums is sorted in ascending order.

```
def binary_search(nums, target):
    left = 0
    right = len(nums) - 1
    while left <= right:
        mid = left + (right - left) // 2
        if nums[mid] == target:
            return mid
        elif nums[mid] < target:
            left = mid + 1
        else:
            right = mid - 1
    return -1

nums = [-1, 0, 3, 5, 9, 12]
target = 9
result = binary_search(nums, target)
print(result)
```

4